Data Quality Assessment: Hospital Operations & Financial Intelligence

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1. Purpose and scope

This document records the checks run on the hospital dataset before the Power BI dashboard was built, what each check found and how each issue was handled. The data covers one hospital's encounters, patients, procedures and payers from 2 January 2011 to 5 February 2022. No source values were deleted or overwritten. Questionable records were kept and marked with helper columns, so the effect of each issue on every KPI can be shown.

2. Data sources

File	Rows	Columns	Role in the model
encounters.csv	27,891	14	Central fact table
procedures.csv	47,701	9	Second fact table, linked through the encounter
patients.csv	974	20	Dimension (identifying columns removed)
payers.csv	10	7	Dimension
organizations.csv	1	8	Dimension (one hospital)
data_dictionary.csv	n/a	3	Reference only, not loaded into the model

3. Summary of findings

ID	Area	Finding	Evidence	Severity	Decision
DQ1	Keys	All relationships resolve	No orphan patient, payer or organization keys in encounters; no orphan encounter keys in procedures; all 974 patients have at least one encounter	Pass	Relationships built as one-to-many
DQ2	Duplicates	No duplicate rows	0 duplicate rows in any table; encounter Id is unique	Pass	None needed
DQ3	Dictionary	Dictionary lists fields the patients file does not	Middle and FIPS County Code are documented but absent	Low	Ignored; neither is used in any KPI

ID	Area	Finding	Evidence	Severity	Decision
		contain			
DQ4	Missing values	Blank optional fields	See section 4.5	Low	Left blank; none feeds a KPI
DQ5	Timestamps	START and STOP are UTC ISO timestamps	Format yyyy-MM-ddTHH:mm:ssZ	Low	Converted to date-time; analysed in UTC as recorded
DQ6	Coverage period	2022 is a partial year	220 encounters, 1 January to 5 February 2022	Medium	Kept; flagged on the dashboard; not compared with full years
DQ7	Duration	Extreme stay lengths distort the average	75 encounters over 24 hours; 13 over 30 days; 4 over a year; longest 44,930 hours	High	Kept and flagged (Over24h); median reported beside the mean
DQ8	Cost	Encounters with a zero total claim cost	89 encounters (0.32%), each with a positive base encounter cost	Medium	Kept and flagged (CostIssueFlag); average shown with and without them
DQ9	Coverage	Zero payer coverage mixes two situations	13,586 encounters (48.71%): 8,807 NO_INSURANCE and 4,779 with an insurer	Medium	Reported as two groups, not one
DQ10	Procedures	One code can carry several descriptions	4 codes have more than one description	Low	Procedures grouped by description
DQ11	Procedures	Many encounters have no procedure rows	13,221 encounters (47.4%); 14,670 have at least one	Medium	Procedure KPIs state their base
DQ1	Patients	Encounters	53 encounters for 52	Low	Kept; 0.19% of

ID	Area	Finding	Evidence	Severity	Decision
2		dated after the recorded death date	patients, up to 14 days after; 50 wellness, 3 emergency		encounters
DQ13	Scope	Adult cohort, single hospital	Patient age at encounter ranges from 19.9 to 99.3 years; one organization	Low	Findings apply to this adult cohort and this hospital only
DQ14	Privacy	Patient file holds personal identifiers	Names and addresses present in the source	Handled	Removed in Power Query; excluded from the repository and screenshots

4. Findings that affect the KPIs

4.1 Duration outliers (DQ7)

The average encounter lasts 7.27 hours, but the median is only 0.25 hours (15 minutes) and the average excluding encounters over 24 hours is 1.54 hours. A few records run for months or years, the longest being 44,930 hours, about 5.1 years. Of the 75 encounters over 24 hours, 58 are inpatient, 7 outpatient, 5 emergency and 5 ambulatory. These may be genuine long-running episodes or recording artefacts, and the data alone cannot say which. The mean is therefore not a reliable measure of a typical stay. The dashboard therefore shows the median next to it, and reports the over-24-hour share (0.27%) separately.

4.2 Zero total claim cost (DQ8)

Eighty-nine encounters have a total claim cost of exactly zero: 51 wellness, 28 inpatient, 6 emergency and 4 ambulatory. None is negative. Each still has a positive base encounter cost, so the zero looks like a missing charge rather than a free visit. The overall average claim cost is 3,639.68; excluding these records it is about 3,651, a difference of roughly 0.3%, so the issue is small but visible.

4.3 Zero payer coverage (DQ9)

Payer coverage is zero for 13,586 encounters (48.71%). Of these, 8,807 belong to the NO_INSURANCE payer and 4,779 belong to a named insurer that recorded no coverage. The two groups mean different things: the first is lack of insurance, the second is unexplained and may reflect denied or unrecorded claims. No coverage value exceeds its claim cost and none is negative. Overall, payers cover 30.63% of total claim cost.

4.4 Partial year 2022 (DQ6)

The first encounter is dated 2 January 2011 and the last 5 February 2022, so 2011 is close to a full year and 2022 holds only five weeks of data. Year-on-year comparisons against 2022 would show a false collapse. The dashboard keeps 2022 visible with a note rather than removing it, so totals still agree with the headline figure of 27,891 encounters.

4.5 Missing values (DQ4)

Table	Field	Blank count	Share	Treatment
encounters	REASONCODE, REASONDESCRIPTION	19,541	70.1%	Left blank; reasons not analysed
procedures	REASONCODE, REASONDESCRIPTION	36,945	77.5%	Left blank; reasons not analysed
patients	DEATHDATE	820	84.2%	Expected for living patients
patients	SUFFIX	953	97.8%	Not used
patients	MAIDEN	588	60.4%	Not used
patients	ZIP	142	14.6%	Not used
patients	MARITAL	1	0.1%	Not used
payers	Address, city, state, zip, phone	1 payer	10%	The NO_INSURANCE row; kept

5. Assumptions

- Each row in encounters is one encounter, and each row in procedures is one procedure performed during an encounter.
- Timestamps are analysed in UTC exactly as recorded, with no conversion to local time.
- Zero payer coverage on a NO_INSURANCE encounter means uninsured. Zero coverage with a named insurer is treated as unexplained.
- The 89 zero-cost encounters and the 75 long-duration encounters are real records, so they stay in the data and are flagged.
- Unique Patients counts distinct patients appearing in the encounters being viewed, not all patients on file.
- In the top-10 procedure ranking by average base cost, a procedure needs at least 20 records to qualify. The threshold is a judgement: hemodialysis (27 records, average about 29,300) just clears it and would drop out at a threshold of 50.
- Ranking procedures by description rather than code avoids splitting counts across the four codes with several descriptions.

6. Transformations applied

1. Loaded the five CSV files and set data types (dates, decimals, text keys).
2. Added StartDate (the date part of START) so encounters can link to the date table.
3. Added DurationHours (STOP minus START, in hours) and the flag Over24h.
4. Added the flags ZeroCoverage (payer coverage equals zero) and CostIssueFlag (total claim cost of zero or less).
5. Added DurationBand (Over 24 hours or 24 hours or less) for charting.
6. In patients, kept only Id, BIRTHDATE, DEATHDATE, MARITAL, RACE, ETHNICITY, GENDER, CITY, STATE and COUNTY. All name, address and other identifying columns were removed.
7. In procedures, kept ENCOUNTER, CODE, DESCRIPTION and BASE_COST.
8. Built a date table from 1 January 2011 to 31 December 2022, with Year, Quarter, Month and YearQuarter columns, and marked it as the date table.
9. Created one-to-many relationships from patients, payers, organizations and the date table to encounters, and from encounters to procedures.

7. Limitations

- One hospital and an adult-only population: the results cannot be generalised to other hospitals or to children.
- No source field explains why coverage is zero for an insured patient, why a claim cost is zero, or why a few encounters last years.
- Procedure analysis covers only the 52.6% of encounters that have procedure rows.
- Readmission analysis was not attempted, because no field defines a clinical readmission and treating every later encounter as one would overstate it.

8. Recommended follow-ups

- Keep the Median Duration Hours card beside the average on the Encounter Patterns page, and mention the 75 long-stay encounters wherever the average is shown.
- Ask the data owner why 4,779 insured encounters show zero payer coverage and why 89 encounters have a zero claim cost.
- Treat any conclusion that rests on 2022 as provisional until the full year is available.